

Foundations of Receiver operating characteristic

[Receiver operating characteristic]

1.0 Content Block

Balakrishnan, Narayanaswamy (1991); Handbook of the Logistic Distribution, Marcel Dekker, Inc., ISBN 978-0-8247-8587-1 Brown, Christopher D.; Davis, Herbert T. (2006). "Receiver operating characteristic curves and related decision measures: a tutorial". *Chemometrics and Intelligent Laboratory Systems*. 80: 24–38. doi:10.1016/j.chemolab.2005.05.004. Rotello, Caren M.; Heit, Evan; Dubé, Chad (2014). "When more data steer us wrong: replications with the wrong dependent measure perpetuate erroneous conclusions" (PDF). *Psychonomic Bulletin & Review*. 22 (4): 944–954. doi:10.3758/s13423-014-0759-2. PMID 25384892. S2CID 6046065. Fawcett, Tom (2004). "ROC Graphs: Notes and Practical Considerations for Researchers" (PDF). *Pattern Recognition Letters*. 27 (8): 882–891. CiteSeerX 10.1.1.145.4649. doi:10.1016/j.patrec.2005.10.012. Gonen, Mithat (2007); *Analyzing Receiver Operating Characteristic Curves Using SAS*, SAS Press, ISBN 978-1-59994-298-8 Green, William H., (2003) *Econometric Analysis*, fifth edition, Prentice Hall, ISBN 0-13-066189-9 Heagerty, Patrick J.; Lumley, Thomas; Pepe, Margaret S. (2000). "Time-dependent ROC Curves for Censored Survival Data and a Diagnostic Marker". *Biometrics*. 56 (2): 337–344. Bibcode:2000Biome..56..337H. doi:10.1111/j.0006-341x.2000.00337.x. PMID 10877287. S2CID 8822160. Hosmer, David W.; and Lemeshow, Stanley (2000); *Applied Logistic Regression*, 2nd ed., New York, NY: Wiley, ISBN 0-471-35632-8 Lasko, Thomas A.; Bhagwat, Jui G.; Zou, Kelly H.; Ohno-Machado, Lucila (2005). "The use of receiver operating characteristic curves in biomedical informatics". *Journal of Biomedical Informatics*. 38 (5): 404–415. CiteSeerX 10.1.1.97.9674. doi:10.1016/j.jbi.2005.02.008. PMID 16198999. Mas, Jean-François; Filho, Britaldo Soares; Pontius, Jr, Robert Gilmore; Gutiérrez, Michelle Farfán; Rodrigues, Hermann (2013). "A suite of tools for ROC analysis of spatial models". *ISPRS International Journal of Geo-Information*. 2 (3): 869–887. Bibcode:2013IJGI....2..869M. doi:10.3390/ijgi2030869.{{cite journal}}: CS1 maint: multiple names: authors list (link) Pontius, Jr, Robert Gilmore; Parmentier, Benoit (2014). "Recommendations for using the Relative Operating Characteristic (ROC)". *Landscape Ecology*. 29 (3): 367–382. Bibcode:2014LaEco..29..367P. doi:10.1007/s10980-013-9984-8. S2CID 15924380.{{cite journal}}: CS1 maint: multiple names: authors list (link) Pontius, Jr, Robert Gilmore; Pacheco, Pablo (2004). "Calibration and validation of a model of forest disturbance in the Western Ghats, India 1920–1990". *GeoJournal*. 61 (4): 325–334. Bibcode:2004GeoJo..61..325G. doi:10.1007/s10708-004-5049-5. S2CID 155073463.{{cite journal}}: CS1 maint: multiple names: authors list (link) Pontius, Jr, Robert Gilmore; Batchu, Kiran (2003). "Using the relative operating characteristic to quantify certainty in prediction of location of land cover change in India". *Transactions in GIS*. 7 (4): 467–484. Bibcode:2003TrGIS...7..467P. doi:10.1111/1467-9671.00159. S2CID 14452746.{{cite journal}}: CS1 maint: multiple names: authors list (link) Pontius, Jr, Robert Gilmore; Schneider, Laura (2001). "Land-use change model validation by a ROC method for the Ipswich watershed, Massachusetts, USA". *Agriculture, Ecosystems & Environment*. 85 (1–3): 239–248. doi:10.1016/S0167-8809(01)00187-6.{{cite journal}}: CS1 maint: multiple names: authors list (link) Stephan, Carsten; Wesseling, Sebastian; Schink, Tania; Jung, Klaus (2003). "Comparison of Eight Computer Programs for Receiver-Operating Characteristic Analysis". *Clinical Chemistry*. 49 (3): 433–439. doi:10.1373/49.3.433. PMID 12600955. Swets, John A.; Dawes, Robyn M.; and Monahan, John (2000); *Better Decisions through Science*, Scientific American, October, pp. 82–87 Zou, Kelly H.; O'Malley, A. James; Mauri, Laura (2007). "Receiver-operating characteristic analysis for evaluating diagnostic tests and predictive models". *Circulation*. 115 (5): 654–7. doi:10.1161/circulationaha.105.594929. PMID 17283280. Zhou, Xiao-Hua; Obuchowski, Nancy A.; McClish, Donna K. (2002). *Statistical Methods in Diagnostic Medicine*. New York, NY: Wiley & Sons. ISBN 978-0-471-34772-9.

2.0 Content Block

the intercept of the ROC curve with the line at 45 degrees orthogonal to the no-discrimination line - the balance point where Sensitivity = Specificity the intercept of the ROC curve with the tangent at 45 degrees parallel to the no-discrimination line that is closest to the error-free point (0,1) – also called Youden's J statistic and generalized as Informedness the area between the ROC curve and the no-discrimination line multiplied by two is called the Gini coefficient, especially in the context of credit scoring. It should not be confused with the measure of statistical dispersion also called Gini coefficient. the area between the full ROC curve and the triangular ROC curve including only (0,0), (1,1) and one selected operating point (tpr, fpr) – Consistency the area under the ROC curve, or "AUC" ("area under curve"), or A' (pronounced "a-prime"), or "c-statistic" ("concordance statistic"). the sensitivity index d' (pronounced "d-prime"), the distance between the mean of the distribution of activity in the system under noise-alone conditions and its distribution under signal-alone conditions, divided by their standard deviation, under the assumption that both these distributions are normal with the same standard deviation. Under these assumptions, the shape of the ROC is entirely determined by d' . However, any attempt to summarize the ROC curve into a single number loses information about the pattern of tradeoffs of the particular discriminator algorithm.

3.0 Content Block

$$A = \int_{x=0}^1 TPR(FPR - 1(x)) dx = \int_{-\infty}^{\infty} TPR(T) FPR'(T) dT = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} I(T' \geq T) f_1(T') f_0(T) dT' dT = P(X_1 \geq X_0) \quad \{\displaystyle \begin{aligned} A &= \int_{x=0}^1 \{\mathbb{E}[TPR(FPR - 1(x))]\} dx \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} I(T' \geq T) f_1(T') f_0(T) dT' dT \\ &= P(X_1 \geq X_0) \end{aligned}$$

4.0 Content Block

$$P_D = \frac{1}{2} \operatorname{erfc} \left(\frac{\operatorname{erfc}^{-1}(2P_{FA}) - X}{\sqrt{2}} \right) \quad \{\displaystyle P_D = \frac{1}{2} \operatorname{erfc} \left(\frac{\operatorname{erfc}^{-1}(2P_{FA}) - X}{\sqrt{2}} \right)}$$

5.0 Content Block

A classification model (classifier or diagnosis) is a mapping of instances between certain classes/groups. Because the classifier or diagnosis result can be an arbitrary real value (continuous output), the classifier boundary between classes must be determined by a threshold value (for instance, to determine whether a person has hypertension based on a blood pressure measure). Or it can be a discrete class label, indicating one of the classes. Consider a two-class prediction problem (binary classification), in which the outcomes are labeled either as positive (p) or negative (n). There are four possible outcomes from a binary classifier. If the outcome from a prediction is p and the actual value is also p, then it is called a true positive (TP); however if the actual value is n then it is said to be a false positive (FP). Conversely, a true negative (TN) has occurred when both the prediction outcome and the actual value are n, and a false negative (FN) is when the prediction outcome is n while the actual value is p. To get an appropriate example in a real-world problem, consider a diagnostic test that seeks to determine whether a person has a certain disease. A false positive in this case occurs when the person tests positive, but does not actually have the disease. A false negative, on the other hand, occurs when the person tests negative, suggesting they are healthy, when they actually do have the disease. Consider an experiment from P positive instances and N negative instances for some condition. The four outcomes can be formulated in a 2x2 contingency table or confusion matrix, as follows: